Toward Vision-Guided Pick-and-Place for Tabletop Humanoid Manipulation

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Outline

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Robotics Research Setting

- Robotics research is typically developed in stages, beginning in simulation and controlled environments before deployment on physical hardware [KBP13, TFR+17].
- This progression is common because simulation enables faster experimentation, safer testing, and easier debugging than direct real-robot development [KBP13, TFR+17].
- However, robotic systems are often studied under predefined objects, layouts, and task assumptions rather than in open-ended human environments [KMvB+21, ZYS+25].
- As a result, strong performance in controlled settings does not necessarily imply robustness in longer-term deployment [TFR+17, KMvB+21].

Visuomotor Manipulation

- A central problem in robotics is **visuomotor manipulation**: using perception to guide action for tasks such as reaching, grasping, and placing objects [SMF22, SMF23, ZYS+25].
- For humanoid robots, this requires detecting an object, estimating grasp-relevant information, and executing manipulation reliably [ZYS+25].
- Prior work has shown strong progress in visuomotor manipulation, but many systems still assume fixed perception and stable deployment conditions [SMF22, SMF23, BBC+23, GWP+24].
- This makes visuomotor manipulation a useful entry point for studying how perception and action should interact in more adaptive robotic systems [ZYS+25, KMvB+21].

Why Fixed Perception Is a Limitation

- In real deployment, environments change over time: layouts vary, uncertainty affects control, and previously unseen objects can appear [KMvB+21, ZYS+25].
- Fixed perception modules are therefore a limitation for long-term autonomy, because deployed robots may need to recognize new objects after initial training [KMvB+21, ZWY+24].
- Updating perception is itself difficult, since learning new classes can degrade previously learned knowledge through catastrophic forgetting [MC89, Fre99, LH16].
- In robotics, this problem is especially important because perception errors or unstable updates can directly affect downstream manipulation [KMvB+21, MBY+25, ZYS+25].

Research Motivation and Scope

Long-term research vision

Develop an integrated visuomotor system that enables a humanoid robot to:

1. perceive a scene,
2. recognize a target object,
3. adapt object knowledge over time, and
4. execute robust grasp-and-place behavior.

Scope of this study

Controlled tabletop experiments on the Fourier GR-1, with emphasis on continual perception and a manipulation foundation for future perception–action integration.

Prior Work in Visuomotor Manipulation

Prior Work	Key Contribution	Limitation
CLIPort [SMF22]	Semantic grounding with spatially guided manipulation	Assumes fixed perception during task execution
PerAct [SMF23]	Structured 3D action representations for multi-task manipulation	Mostly evaluated with fixed object categories and benchmark setups
RT-1 / Octo / Diffusion Policy [BBC+23, GWP+24, CXF+25]	Large-scale policy learning improves robustness and generalization	Continual adaptation after deployment is not a central focus

Takeaway: Visuomotor manipulation has advanced strongly, but most systems still rely on fixed perception and stable deployment settings.

Prior Work in Continual Perception

Prior Work	Key Contribution	Limitation
LwF / iCaRL [LH16, RKSL17]	Incremental learning methods can reduce catastrophic forgetting	Mostly validated on vision benchmarks rather than robot behavior
LoRA [HSW+22]	Efficient adaptation through lightweight parameter updates	Efficient updating does not by itself ensure stable downstream manipulation
Lifelong Robotics [KMvB+21, MBY+25]	Long-term autonomy requires continuous learning over time	Practical coupling between adaptive perception and action remains limited

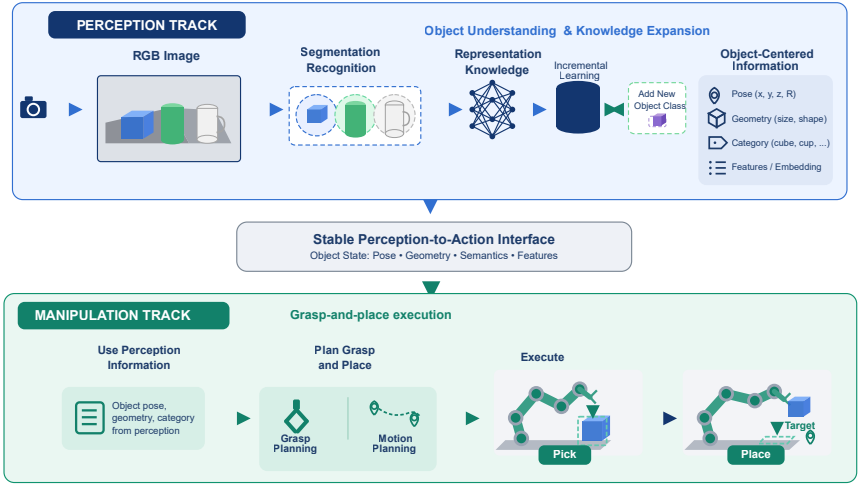
Takeaway: Continual learning supports adaptive perception, but embodied deployment is still much less mature.

Gap at the Perception–Action Interface

Research Line	What It Establishes	Open Gap
Visuomotor manipulation [SMF23, BBC+23, GWP+24]	Strong task execution and learned policy behavior	Usually assumes fixed perception and stable object knowledge
Continual perception [HSW+22, RKSL17, ZWY+24]	Incremental knowledge update with reduced forgetting	Rarely evaluated through downstream grasping or placement
Embodied lifelong robotics [KMvB+21, MBY+25]	Perception and action should evolve together for long-term autonomy	A stable perception-to-action interface is still missing

Takeaway: The central gap is not perception or manipulation alone, but stable coordination between evolving perception and downstream action.

Overall System



Current Methodological Decomposition

- The overall research direction is a modular **visuomotor pipeline**.
- In the long term, the **perception track** should update object knowledge over time and provide object-centered information for action.
- The **manipulation track** should use that information to plan and execute grasp-and-place behavior.
- In the current study, these two tracks are investigated separately:
  - a **manipulation study** for tabletop pick-and-place control,
  - a **perception study** for incremental object understanding.

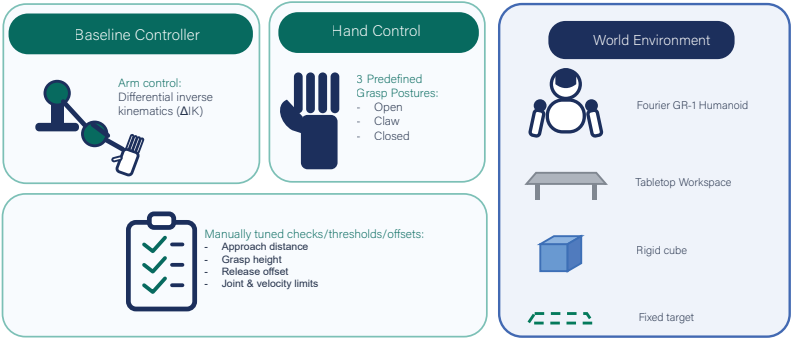
Reason for this decomposition

It isolates the two sides of the perception–action interface before full integration.

Manipulation Study: Objective and Setup

Goal

Develop a simulated tabletop pick-and-place task for the Fourier GR-1, assuming cube position is known.



Manipulation Study: Scripted Baseline

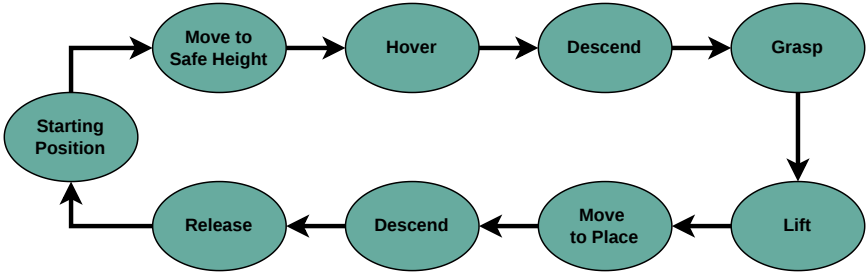


Figure: Hand-designed state machine used as a structured baseline for tabletop pick-and-place.

Main limitation

Relies on fixed assumptions, heuristic thresholds, and manually designed stage logic.

Manipulation Study: RL Reformulation

- To reduce dependence on hand-crafted stage logic, the task is reformulated as a **reinforcement learning problem**.
- Training is performed in a parallel Isaac Lab environment using **PPO**.
- The observation includes end-effector state, cube state, placement target, relative geometry, and arm/hand joint states.
- The action consists of incremental arm joint commands and a scalar grip-control signal.
- The reward encourages reaching, lifting, transport, and successful placement, with penalties for unstable or unnecessary motion.

Methodological role

This reformulation tests where scripted control becomes insufficient and where learned control becomes necessary.

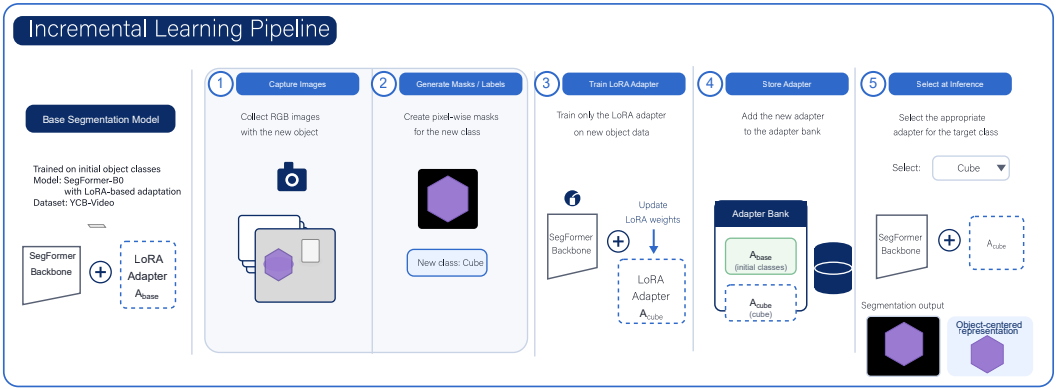
Perception Study: Objective

Goal

Develop an incremental semantic segmentation module that can learn new object classes over time while minimizing catastrophic forgetting.

- The perception track is motivated by open-world deployment, where new objects may appear after initial training.
- The target is not only recognition accuracy, but also stable object knowledge that can later support downstream manipulation.
- The long-term role of this module is to provide object-centered information for a future perception-to-action interface.

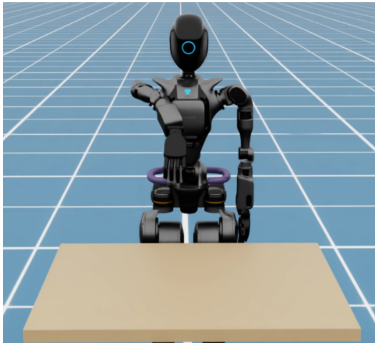
Perception Study: Incremental Pipeline



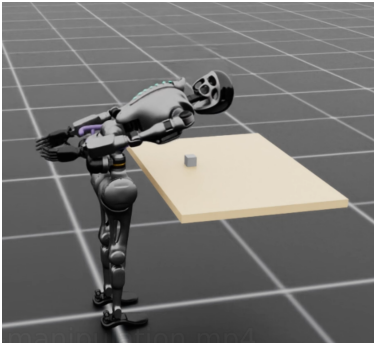
Perception Study: Dataset and Experimental Setup

- **Dataset:** YCB-Video
  - 21 household object classes,
  - RGB-D video with pixel-level annotations,
  - class split into base classes (1–10) and new classes (11–21).
- **Proof-of-concept subset:**
  - 5 sequences,
  - 6,638 usable frames,
  - 5,974 train / 664 test.
- **Model:** SegFormer-B0 with a modified segmentation head.
- **Adaptation mechanism:** LoRA inserted into selected transformer layers.
- **Metric:** mean Intersection-over-Union (mIoU).

Results: Manipulation Study



Manual Pick-and-Place Demo



RL Training Demo

Results: Perception Study

Method	Base IoU Before	Base IoU After	New Class IoU
Naive Fine-Tuning	0.96	0.18	0.95
LoRA Fine-Tuning	0.96	0.19	0.93
Adapter Fusion	0.96	0.97	0.42
LoRA + Distillation	0.96	0.8833	0.9144

Takeaway: LoRA + distillation gives the best retention–plasticity trade-off in the current setup.

Current Status Across Both Tracks

Track	Capability	Level 0	Level 1	Level 2	Level 3	Level 4
Manipulation		*Hard-coded*	*Reactive*	*Adaptive*	*Autonomous*	*Dynamic*
	Grasp Planning	None	Model-based (Stage 2)	Model-free isolated (Stage 3)	Model-free cluttered	Articulated / deformable
	Execution	Hard-coded (Stage 1)	Reactive single-modal	Reactive multi-modal (Stage 3)	Autonomous under uncertainty (Target)	Autonomous dynamic
Perception		*Static*	*Offline Incremental*	*Retention-Aware*	*Online Adaptive*	*Embodied Continual*
	Object Knowledge	Fixed class set	Offline class-incremental updates	LoRA + distillation retention	Online novel-class expansion	Continual open-world object learning
	System Integration	Standalone segmentation	Dataset-level evaluation	Incremental method comparison	On-robot perception updates	Perception-action integration (Target)

Takeaway: Manipulation is moving toward learned control; perception has reached offline incremental learning and now points toward embodied integration.

Next Steps

- On the perception side, move from offline incremental segmentation toward online novel-class adaptation and robot-level integration [KMvB+21, MBY+25].
- On the manipulation side, improve the RL policy until grasping is more robust to pose variation.
- Reduce reliance on fixed offsets and predefined stage assumptions in manipulation.
- Integrate the perception and manipulation modules into a single visuomotor pipeline.
- After stable simulation performance, transfer the integrated system to the physical Fourier GR-1 robot.

Conclusion

- This work targets a foundational robotics problem: tabletop perception-to-manipulation with a humanoid robot [ZYS+25].
- The current contribution on the manipulation side is a progression from scripted control to RL-based adaptive control.
- The perception-side contribution is an incremental segmentation study showing that LoRA + distillation gives the strongest retention/plasticity trade-off in the current setup [LH16, HSW+22].
- Together, these studies establish the basis for a larger integrated visuomotor system.

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THANK

QUESTIONS?